

EVOLUTIONARY BIOLOGY:

It is the history of life forms on earth.

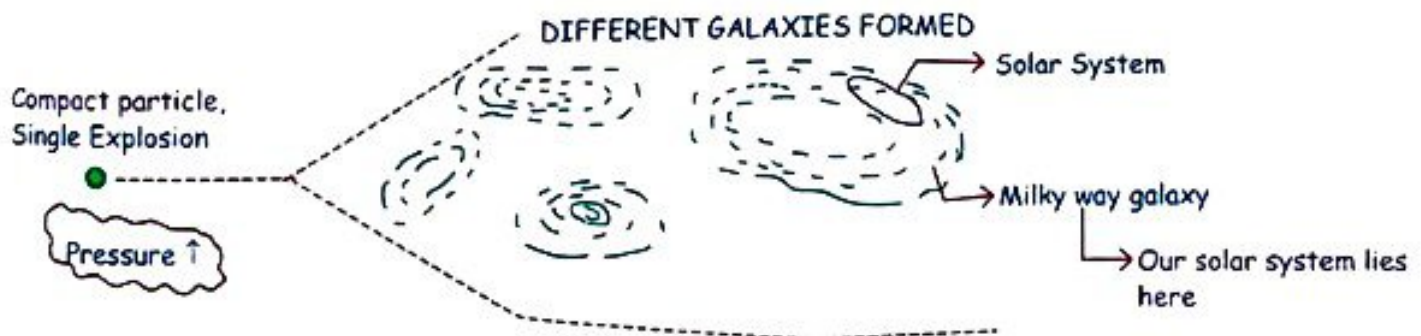
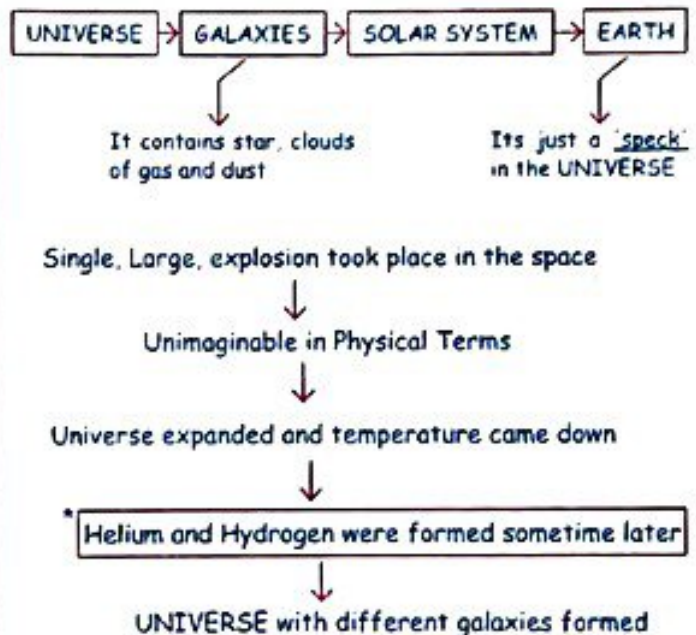
Evolution means to "unfold" or "unroll"

- Changes in the Flora & 'Fauna' that took place gradually over million of years.
- These changes are important for the Survival of the particular organism.

ORIGIN OF UNIVERSE:

BIG BANG THEORY This theory explains the 'ORIGIN OF UNIVERSE'

- Origin of Universe took place around 20 Bya (Billion years ago)

**NOTE**

- 1 light year is the distance travelled by light in one year.
- Stellar distances are measured in light years
- Distance between planets, stars and galaxies are called stellar distances.

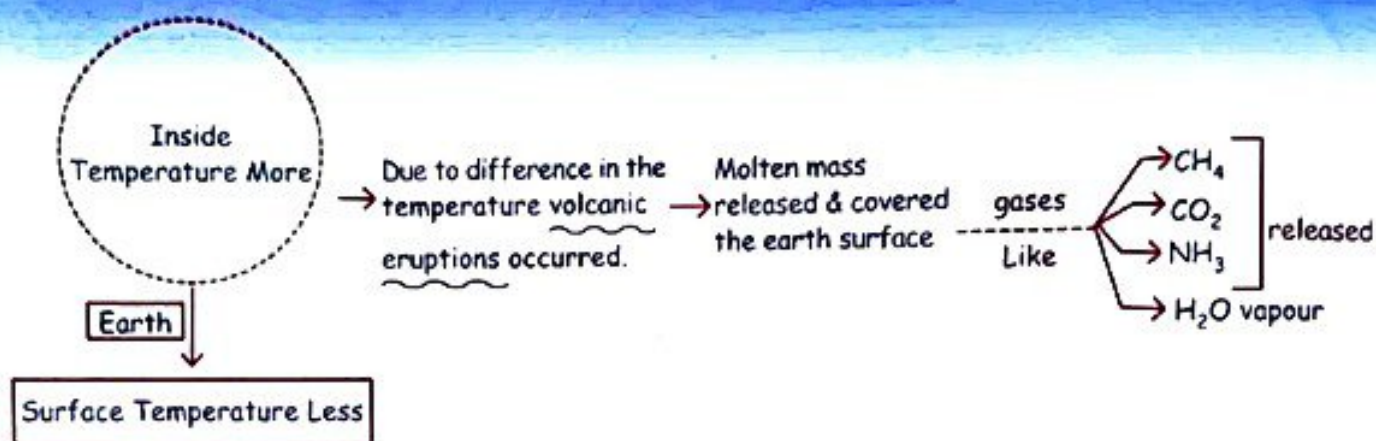
ORIGIN OF EARTH

- 4.5 Billion years ago
- Life appeared 500 million years after formation of earth ~ 4 Billion years ago

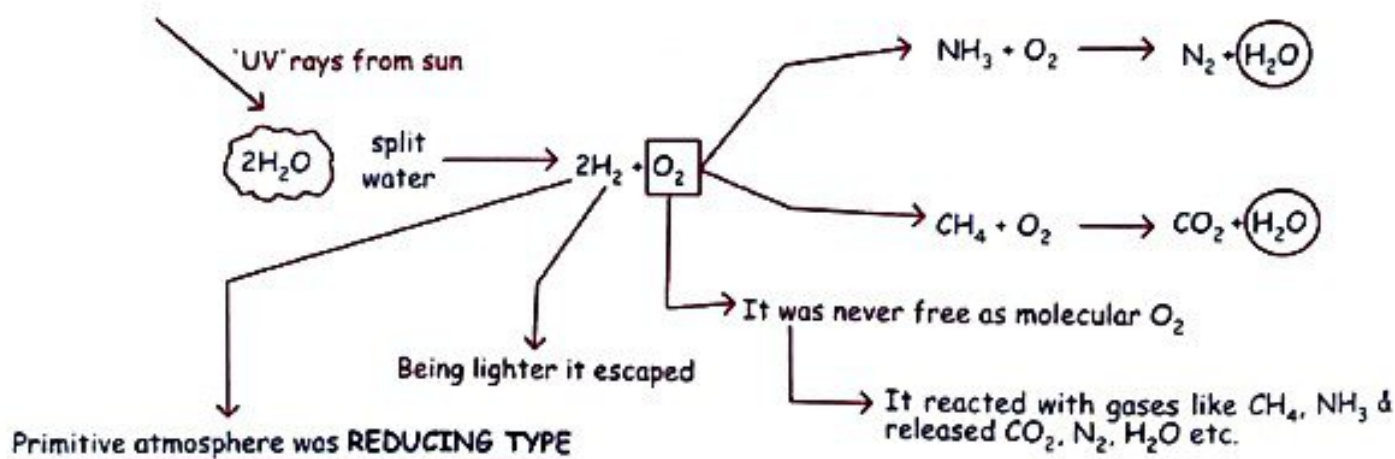
NOTE

- First non-cellular form of life: 3 Billion years ago
- First cellular form of life: 2 Billion years ago

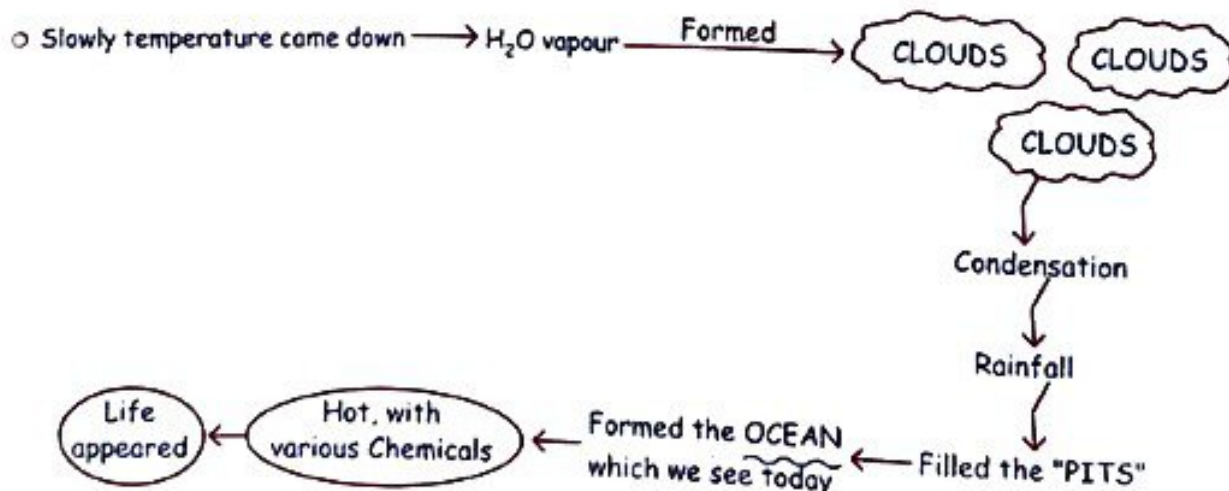
ORIGIN OF EARTH



- In the primitive atmosphere, there was no 'ozone layer', UV rays could easily enter.



- Later on this O_2 was also responsible for the formation of 'ozone layer'
- Earlier Atmosphere was 'Reducing type', with high temperature.

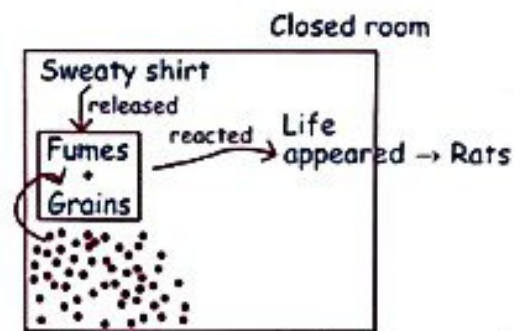


- 1st LIFE APPEARED IN WATER.

THEORIES TO EXPLAIN THE ORIGIN OF LIFE

1. **Theory of special creation:**
 - Life was created within 6 days by the supernatural power called as GOD.
 - Life is created 'as such'. Diversity was same since creation and will remain same in the future.
 - Earth is about 4000 years old.
2. (अतिनिवृत्त Gyan) **Theory of Catastrophism:** According to this, due to some catastrophic events, existing life disappears & new life come.
3. **Theory of Panaspermia/Cosmozoic** (Extra terrestrial theory)
Life came on the Earth from other planet in the form of sperm, seed, spore etc.
4. **Theory of Abiogenesis/Spontaneous Generation**

Abiogenesis : Life appears from non-living matters (decaying and rotting material) like straw and mud.
 absent Life → generation



It was believed by some that due to the fumes & grains in the closed room, rats appeared spontaneously.

(😂 मतलब कुछ भी)

5. Theory of BIOGENESIS:

Living Generation

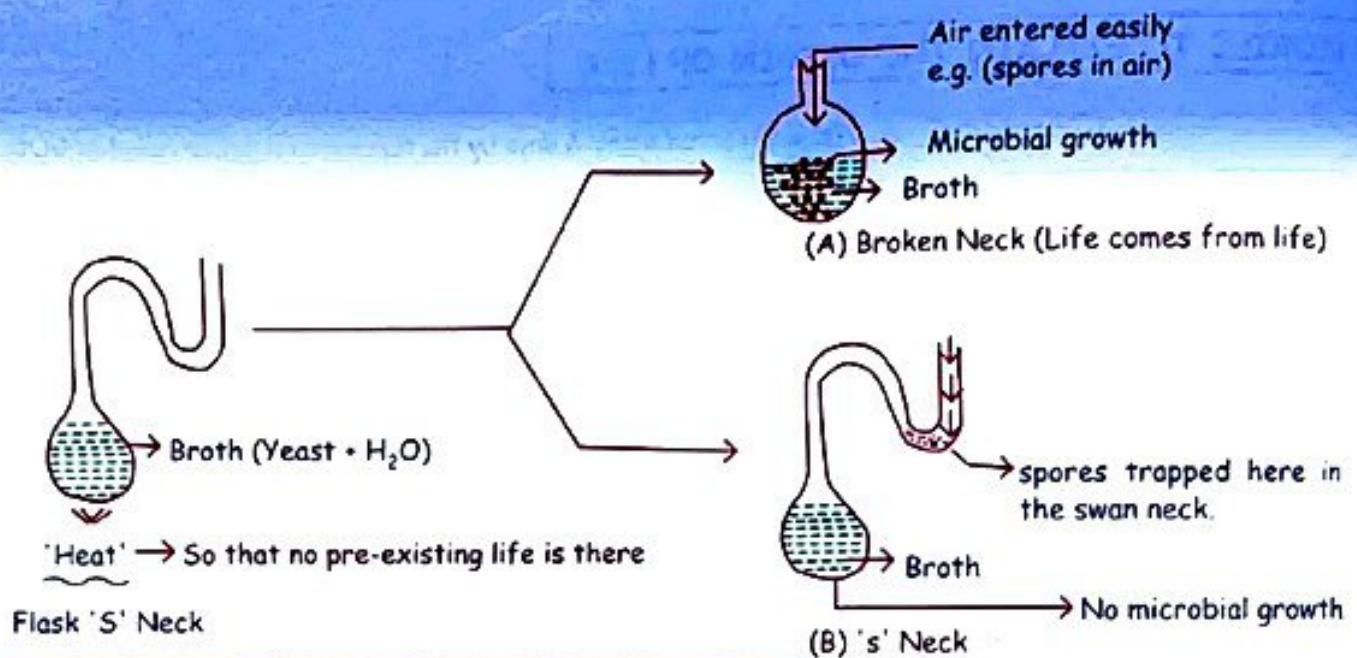
Life comes from pre-existing life (Omne Vivum EX VIVO)

NOTE

- Theory of Biogenesis disproved the theory of abiogenesis.

Pasteur's Swan Neck Experiment:

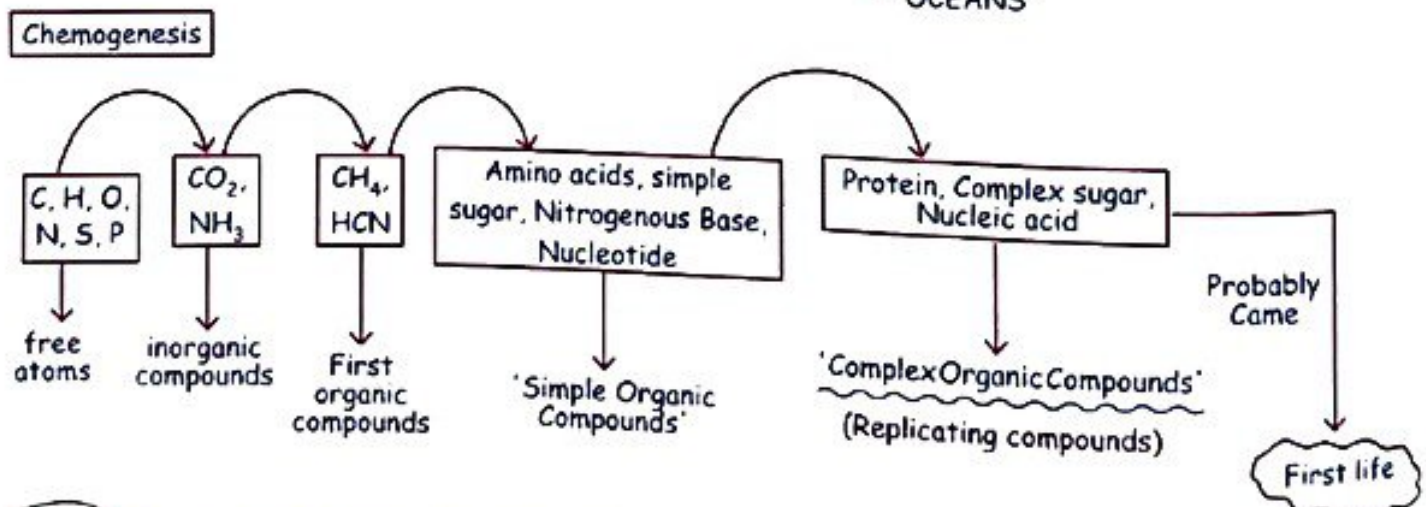
- He took swan 's' neck flask with pre-sterilised condition having broth,
 - (A) : It was broken
 - (B) : It was kept as it 'is'



○ HENCE, PROVED LIFE COMES FROM PRE-EXISTING LIFE.

6. THEORY OF CHEMICAL EVOLUTION (CHEMOGENESIS) : (OPARIN AND HALDANE THEORY):

- Oparin (Russian Biochemist) & Haldane (England Biologist) proposed that the first form of life come from pre existing non-living ORGANIC molecules LIKE RNA, PROTEIN etc.
- Most Accepted Theory
- They proposed that the condition on early earth supported the origin of life like:
 - (i) High temperature
 - (ii) Volcanic eruption
 - (iii) Reducing atmosphere
 - (iv) Lightening
- They said life originated in water, Haldane used the term "hot dilute soup" for 'OCEANS'



NOTE Exactly how the non-replicating components formed replicating is not yet clear.

MILLER & UREY'S EXPERIMENT

- They experimentally proved the theory of 'Chemogenesis' in favour of Oparin & Haldane theory in 1953.

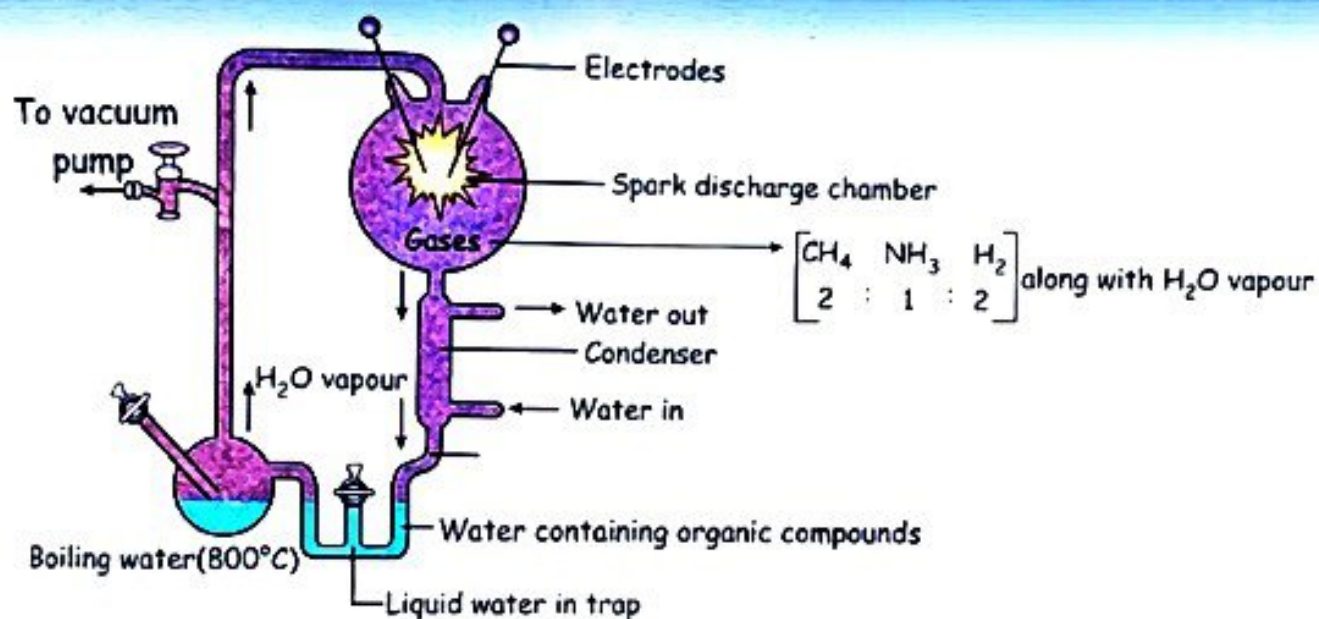


Fig: Diagrammatic representation of Miller's experiment

CONCLUSION

- After 18 days of his experiment, he obtained amino acid like glycine, alanine & aspartic acid.
- Similar experiments were performed by various other scientists to obtain other amino acid, sugar, pigments, fats etc.

NOTE अतिनिकत gyaan

- 1st Life appeared in deep oceans were chemoheterotrophic anaerobes
- Chemoheterotrophs → Chemoautotrophs → Photoautotrophs

Theories of Evolution:

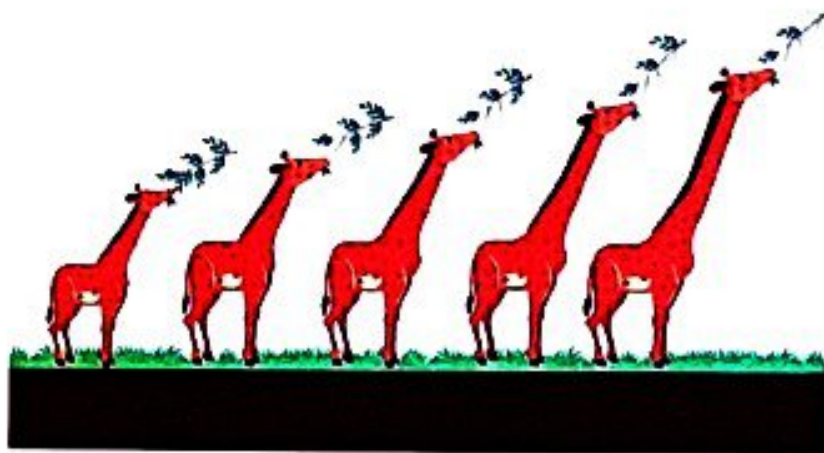
1. LAMARCKISM
 2. DARWINISM
 3. MUTATION THEORY
 4. MODERN SYNTHETIC THEORY
- } 'Biological Evolution'

BIOLOGICAL EVOLUTION:

1. **LAMARCKISM** (Theory of use/disuse of organ or Theory of Inheritance of Acquired Characters)
 - Change in environment force the organism to either develop a new organ or modify the existing organ.
 - The more we use the organ, the more it develops and the less we use it, it disappears.
 - These character that one acquires during the life time is also passed on to the next generation.

- He gave the examples of Giraffes who in an attempt to forage leaves on tall trees had to adapt by elongation of their necks.
- Slowly over years, they came to acquire long necks.

NOTE Lamarck was a French naturalist who gave lamarckism.



अतिनिकत ग्रन्थ

Theory of germplasm by Weismann disproved this theory, he cut the tail of mice for 22 generation, still tails appeared next time.

- Weismann concluded that any change in germplasm of germ cell would pass on to the next generation not the changes in the somatoplasm.

2. **DARWINISM/THEORY OF NATURAL SELECTION**



IMP. NOTE

- Both Darwin & Alfred Wallace were influenced by Population theory of Thomas Malthus.

POPULATION THEORY:

According to this Theory

- Population have a tendency to grow exponentially but it does not as the NATURAL RESOURCES are limited.

- No two individuals are similar, they shows variations most of which are inheritable.
- Size of population is stable except seasonal fluctuation.
- Considering these points, DARWIN PROPOSED DARWINISM which says.
- **Differential reproduction:** Different individual reproduce at different age and different rate.

o Natural resources are limited:

o Struggle for existence "Competition" occurs

- Intraspecific (within species)
- Interspecific (between species)

o *Variation according to Darwin are small and directional → Useful variations are adaptations.

o Natural selection: Nature selects the fittest individual with useful variation which leaves maximum progeny (REPRODUCTIVE FITNESS)

o Survival of the fittest: (Term was given by Herbert Spencer)

NOTE

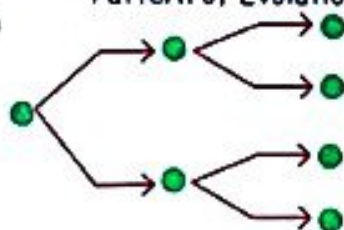
**** 2 KEY CONCEPTS OF DARWINIAN THEORY**

① NATURAL SELECTION

Defines
↓
Mechanism of Evolution

② BRANCHING DESCENT

Defines
↓
Pattern of Evolution



Examples of Natural Selection:

- o Industrial melanism
- o Antibiotic resistant microbes.

- o Darwin Finches (explained later in the chapter)
- o Evolution of pesticide Resistant mosquitoes.

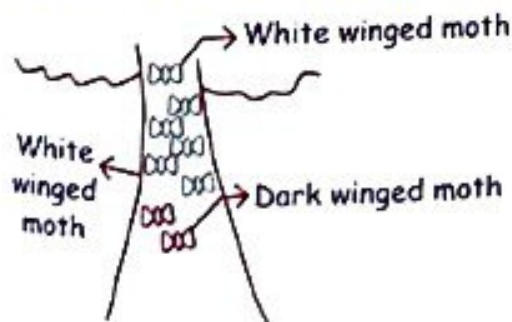
① Industrial Melanism: (England)

Before industrialisation: (1850's)

→ No pollution

↓
Lichens covered the bark of trees before industrialisation (No pollution)

↓
White-winged moth could easily camouflage in the light background whereas the dark-winged moth (melanised) were easily predated by birds

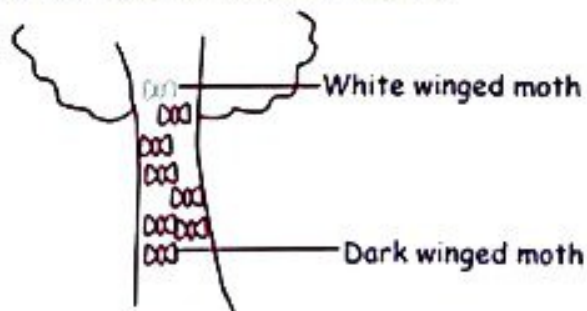


After Industrialisation (1920's)

↓
Pollution increased

↓
Lichens disappeared (lichens don't grow in polluted areas) & due to the pollution trees bark became Dark

↓
White-winged moth easily predated this time & Dark-winged could camouflage





(a)



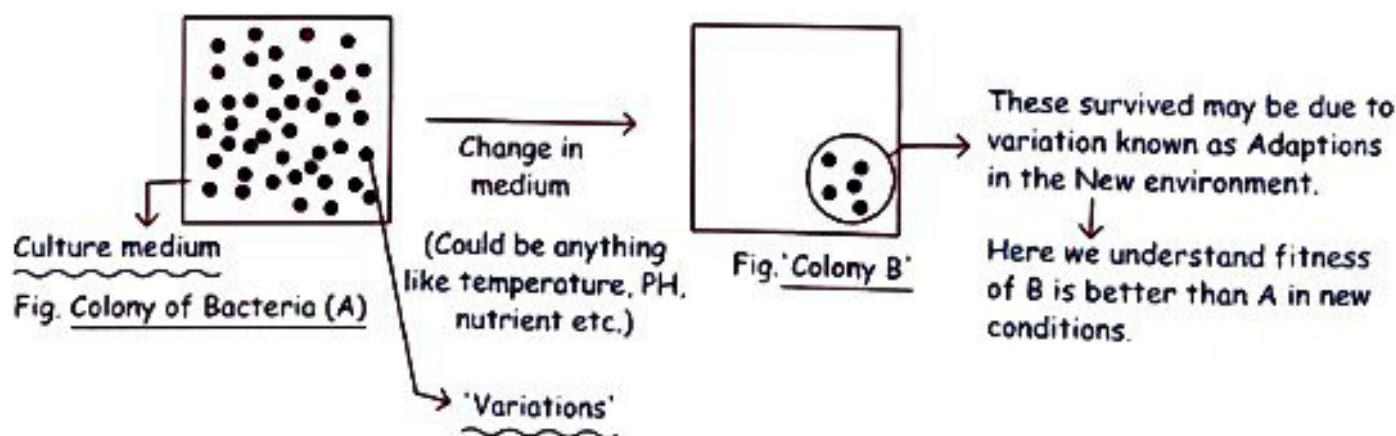
(b)

Fig. Figure showing white - winged moth and dark - winged moth (melanised) on a tree trunk
(a) In unpolluted area (b) In polluted area

NOTE

- Before industrialisation
 - White winged moth increased
 - Dark winged moth decreased
- After industrialisation
 - Dark winged moth increased
 - White winged moth decreased

2. Microbial Experiment:



NOTE The rate of appearance of new life forms is directly linked to the Life span/life cycle of organisms

- Microbial experiment occurs within days but the something to happen for fish or fowl would take million of years as life span of these are in years.

3. Pesticide Resistance Mosquito: Excess of use of herbicides, pesticides has resulted in resistant varieties in much lesser time.

These are examples of evolution by anthropogenic action (human action)

NOTE

Evolution is not a directed process, it is a stochastic (random)

3. Mutation Theory:

- Huger de Vries Proposed this theory
- He worked on 'Evening Primrose'

↓
Oenothera lamarckiana

- He said 'MUTATIONS' results in evolution.

*He gave the term 'SALTATION' for SINGLE STEP LARGE MUTATION: Large differences arising suddenly in a population

- Mutations are Large & Directionless

4. Modern Synthetic Theory:

Many factors collectively are responsible for formation of new species

↓
SPECIATION

- Mutation
- Gene migration
- Genetic drift
- Genetic recombination
- Natural selection

*EVIDENCES OF EVOLUTION:

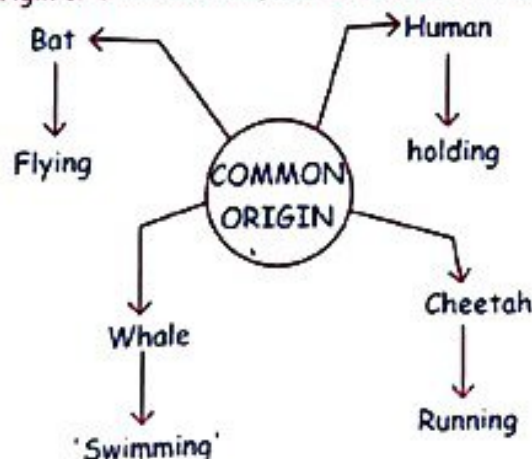
1. Comparative anatomy and morphology
2. Palaeontological evidences
3. Embryological evidences
4. Biochemical evidences

→ HOMOLOGY
→ ANALOGY

1. MORPHOLOGICAL AND ANATOMICAL EVIDENCES:

(a) HOMOLOGOUS STRUCTURE

- Structures having common origin or similar basic plan but differs in functions like forelimbs of vertebrates.



○ Homology explains 'DIVERGENT EVOLUTION'

Even after having similar origin, due to different needs adaptations were different hence they function differently.

Examples:

○ Forelimbs of man, cheetah, whale, bat all have similarities in the pattern of bone like presence of humerus, Radius ulna, carpal, metacarpal, phalanges.

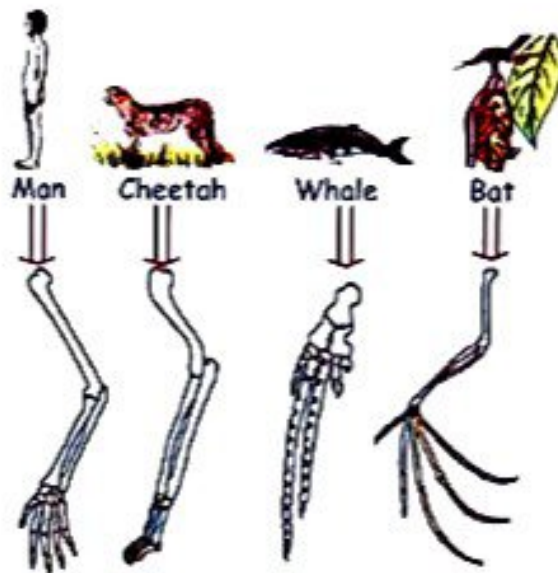


Fig. Example of homologous organ in Animal.

○ Thorns (Protection) of *Bougainvillea* and tendrils (climbing) of *Cucurbita*. Both modified axillary Bud.

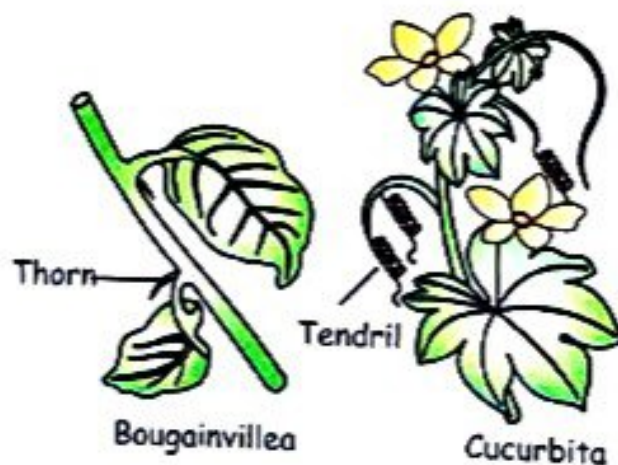


Fig. Example of homologous organ in plant.

○ Heart and Brain of vertebrates

→ Heart of Vertebrates → presence of atria, ventricles, evolution of 2/3/4 Chambered heart
→ Brain of Vertebrates

(b) ANALOGOUS STRUCTURES

- Structure having different origin but common functions.
- Analogy explains 'CONVERGENT EVOLUTION'



Similar habitat that has resulted in selection of similar adaptive features in different groups of organisms but towards the same function.

Examples:

- Wings of bird (Vertebrate) and wings butterfly (Arthropoda): (Flying)
- Eye of Octopus and mammal: (Vision)
- Sweet potato and potato: (Stored food)



Modified Root

Modified shoot

- Flippers of Penguin & Dolphin

2. Palaentological Evidences: (Best Evidence):

Palaentology: Study of FOSSILS

Fossils: Whole organism or its part or excreta or foot print etc. preserved overs years.



found in 'Sedimentary Rocks'

- Most accurate evidence

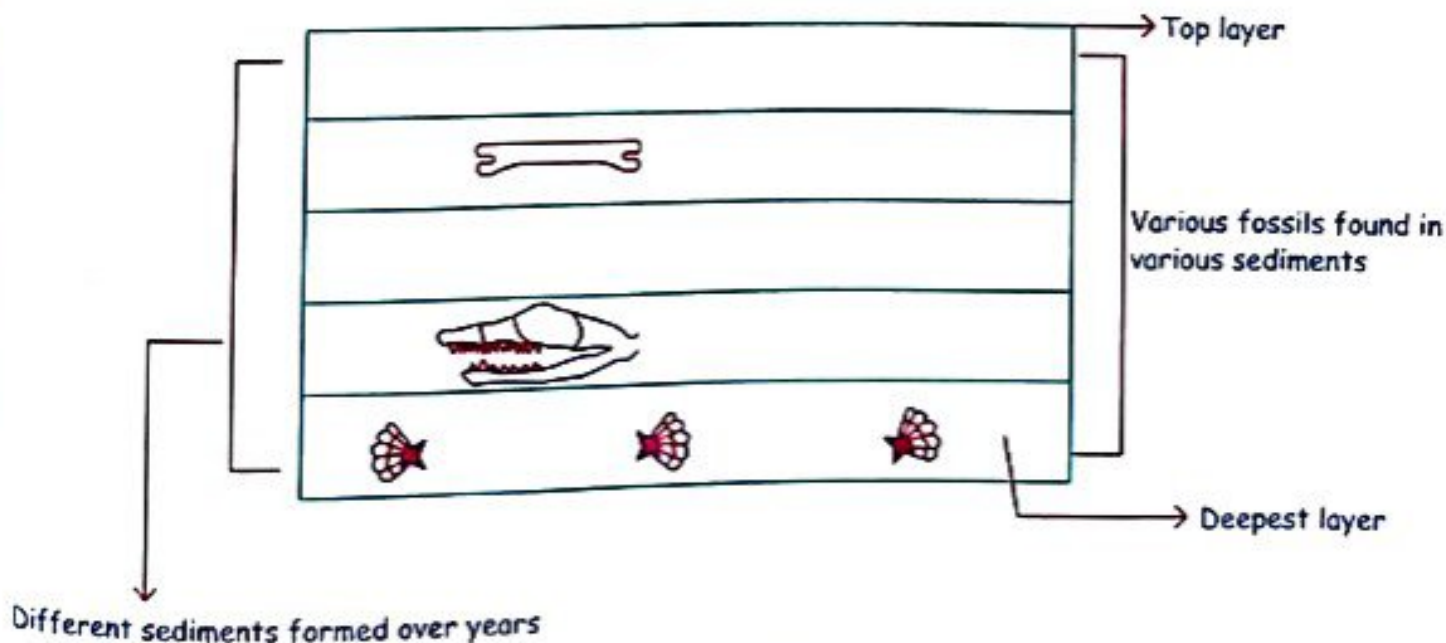


Fig. Different sediments layer over year

○ Rock form sediments one over another.

○ Different aged rock sediment

↓
Contain fossil which died during formation of that particular sediment

○ Deepest layer: OLDEST FOSSIL.

○ Top layer: NEWEST FOSSIL.

NOTE Age of fossils can be determined by:

○ Carbon dating

○ Electron spin resonance

○ Relative dating by comparison of deep & top layers.

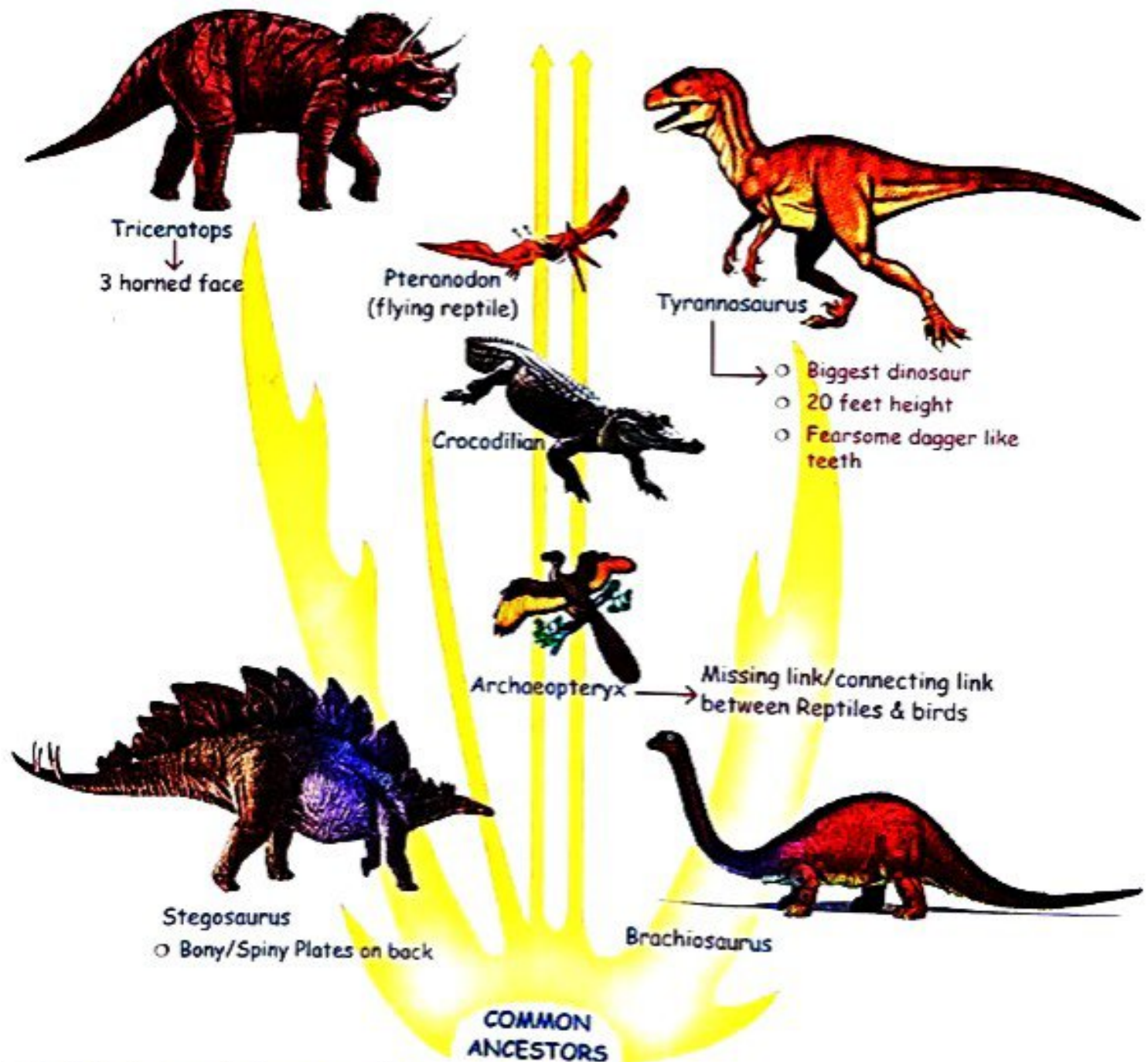


Fig. A family tree of dinosaurs and their living modern day counterpart organisms like crocodiles and birds

3. EMBRYOLOGICAL EVIDENCE:

BIOGENETIC LAW OR RECAPITULATION THEORY

By 'Ernst Haeckel'

ONTOGENY RECAPITULATES PHYLOGENY
Embryonic Stages repeats Evolutionary history

ACCORDING TO BIOGENETIC LAW

- Embryos of all vertebrates have similar structures like gill slits behind head.
gill slits → It is not functional in all adults, only functional in fish adult
→ Indicates COMMON ANCESTRY
- ★ KARL ERNST VON BAER, rejected this theory & said:
- Initial stages common to all but later stages are not common → embryos never passes through adult stages of other animals.

4. BIOCHEMICAL EVIDENCE:

- Similarities in proteins and genes performing a given function among diverse organisms give clues to common ancestry.

e.g:

- Trypsin is common for both vertebrates & invertebrates.

- Apes & human have similar proteins in blood.

ARTIFICIAL SELECTION:

- Man breeds Animals & Plant (Plant breeding and Animal breeding) for different purposes: Agriculture, Horticulture (Fruit & Vegetable), security and sports.
- It is said, if within 100 of years if man could create new breeds why could not nature have done this over million of years.

HW EQUILIBRIUM: (Hardy Weinberg Equilibrium)

Hardy Weinberg principle says that in a large randomly mating population: Allelic frequency remains constant or Stable generation after generation if population is free from:

1. Gene migration
2. Genetic recombination
3. Mutation
4. Natural Selection
5. Genetic drift

Evolutionary forces

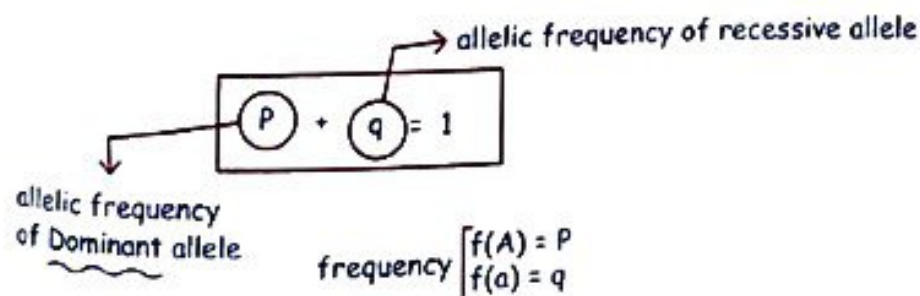
'Gene' alternate form of allele:

'T' : Dominant

't' : Recessive

'genotype' → 'Tt' : Heterozygote
→ 'TT' : Homozygous Dominant
→ 'tt' : Homozygous Recessive

Mathematical Expression for Hardy-Weinberg Equation

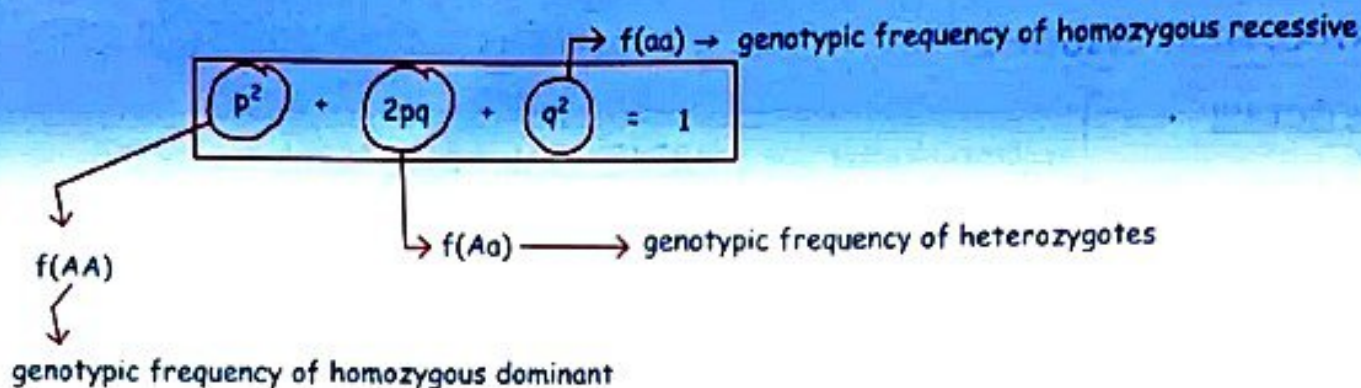


(A) dominant allele (a) recessive allele

(AA) - Homozygous dominant

(Aa) - Heterozygotes

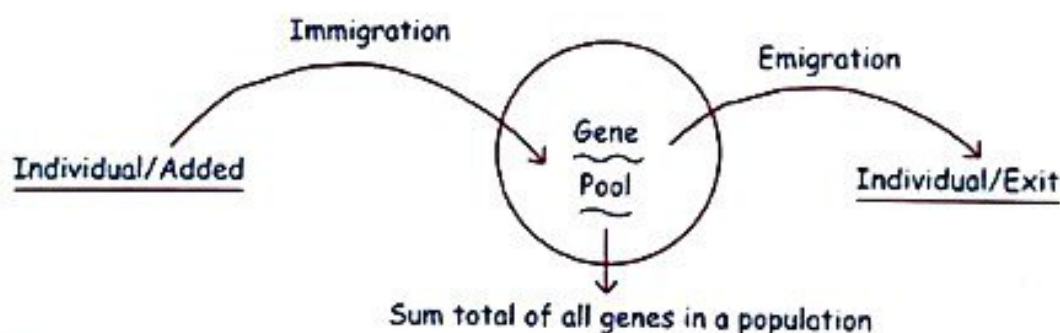
(aa) - Homozygous recessive



NOTE If $p + q \neq 1$, means evolution is taking place in the population & one or more factors are operating.

Factors affecting HW (Hardy-Weinberg) equilibrium

1. Gene flow or Gene Migration:



NOTE

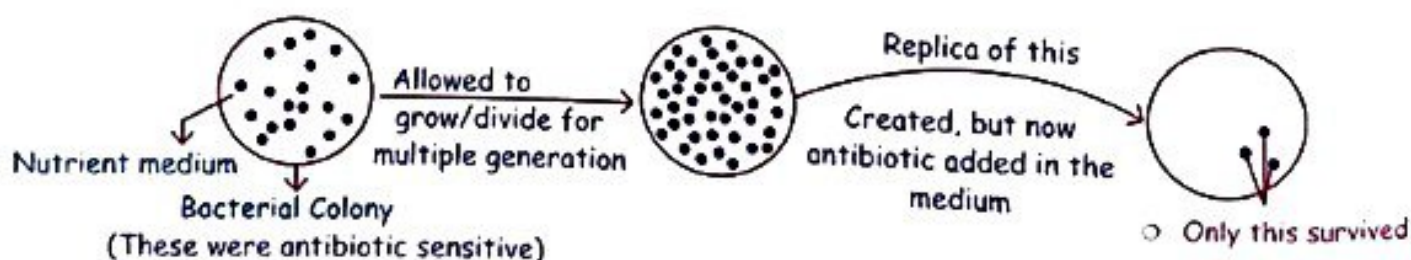
○ If gene migration occurs multiple times: It is known as **GENE FLOW**

2. Genetic recombination: During gamete formation → Recombination occurs during meiosis
 ↓
 Change in gene frequency

3. Mutation:

Lederberg's Experiment:

- Microbial experiment performed to show pre-existing advantageous mutation when selected will result in new phenotypes.
- Over few generations new population will be formed.



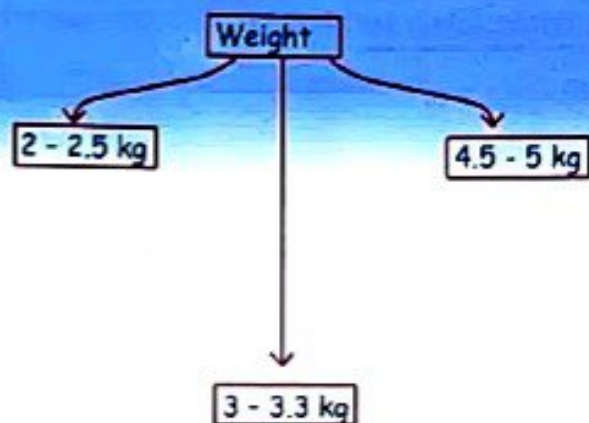
Conclusion: May be over generations, there might be some mutations in the members already but only showed up upon change in environment, will now form new species.

4. Natural Selection (Types of Natural selection)

- ① Normal/Stabilising selection
- ② Progressive/Directional selection
- ③ Disruptive Selection

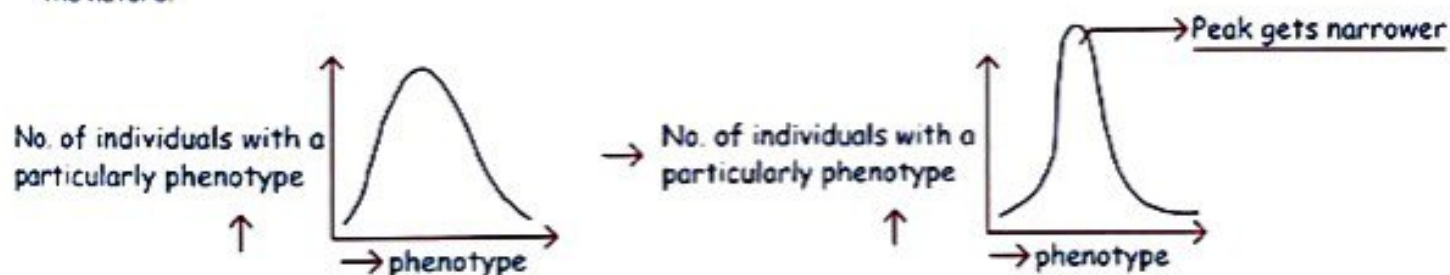
① **Normal/Stabilising Selection:** Nature selects individuals with mean character value.

e.g.: Birth weight of human babies.

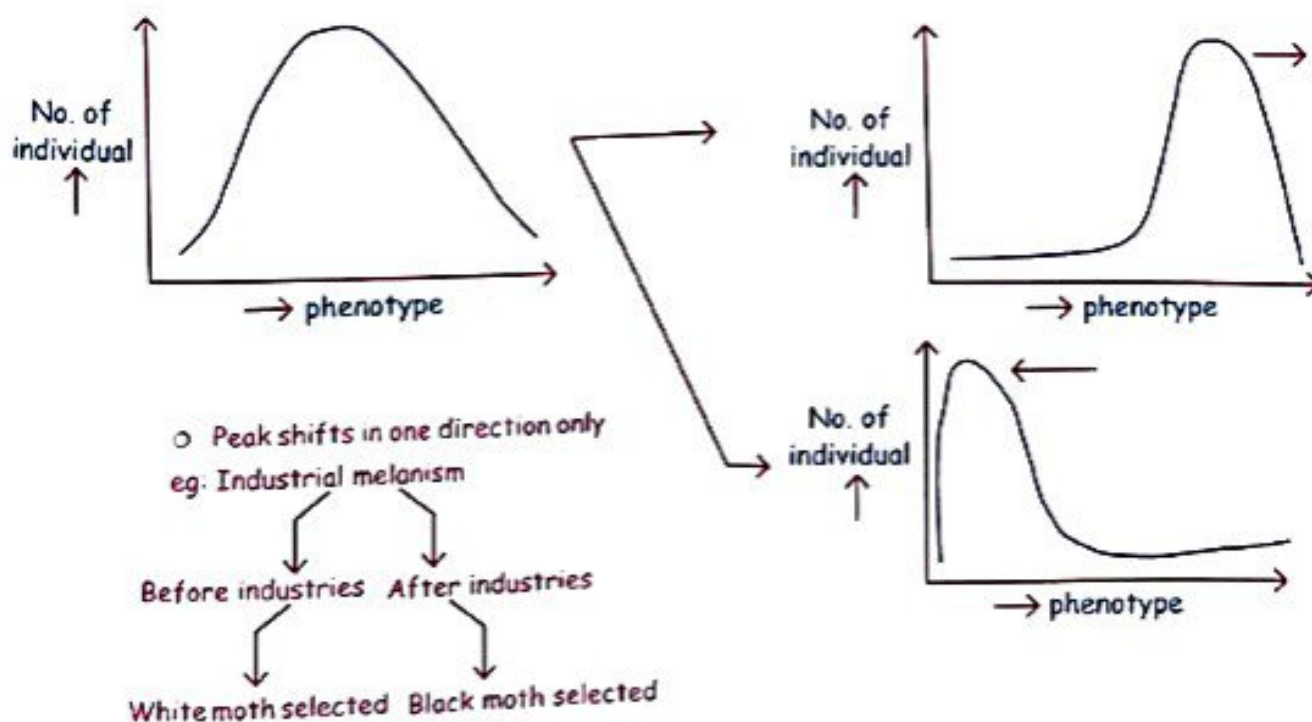


97% of new born with an average weight of this survives as compared to other infants.

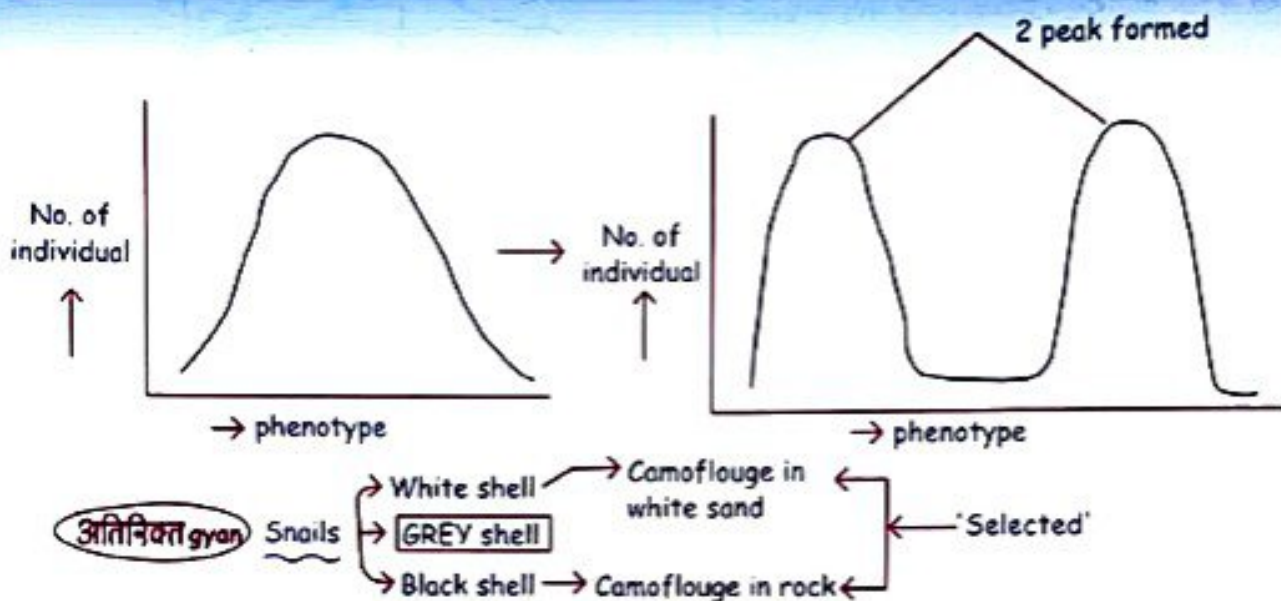
○ In this case, extreme low or extreme high baby weight is not favoured instead Average Baby weight is favoured by the nature.



② **Progressive/Directional Selection:** Nature selects one of the extreme phenotype not average mean value.

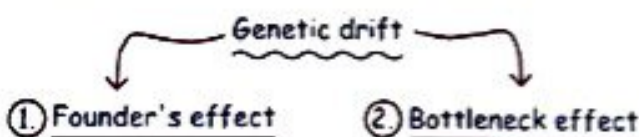


③ **Disruptive selection:** Both peripheral values are selected together.



5. **Genetic Drift:**

- In a small population, random change in gene frequency
- May be due to natural calamity, geographical isolation.



Founder's Effect: By chance, if there is a small group of individual become isolated from the main population, forms a new population: now known as Founder's Population.

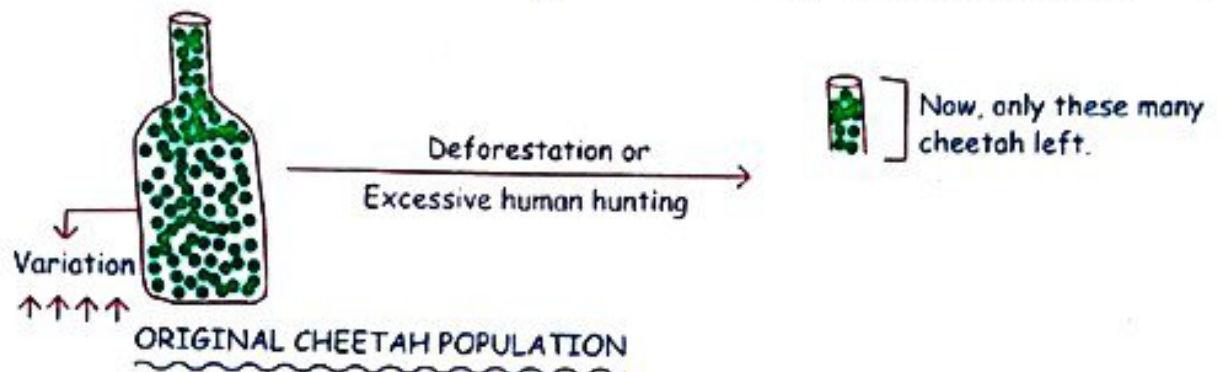
eg: Darwin Finches

So, there was an original seed eating ANCESTOR BIRD in the main land, some of which drifted to a new place & formed a whole New population (species)



Bottleneck Effect: Initially population was enough to appear like filling complete bottle but due to some reason, only a few left which is comparable to the size of neck of the bottle.

e.g.: Excessive human hunting of Cheetah resulted in Bottle neck effect.



ADAPTIVE RADIATION : Evolution of different species in given geographical area starting from a common point.

e.g: Darwin finches, Australian marsupials and placental mammals in Australia.

(i) Darwin finches

Darwin during his sea voyage
↓
H. M. S Beagle
↓ went to
GALAPAGOS ISLAND
↓
Darwin observed small Black Birds over there

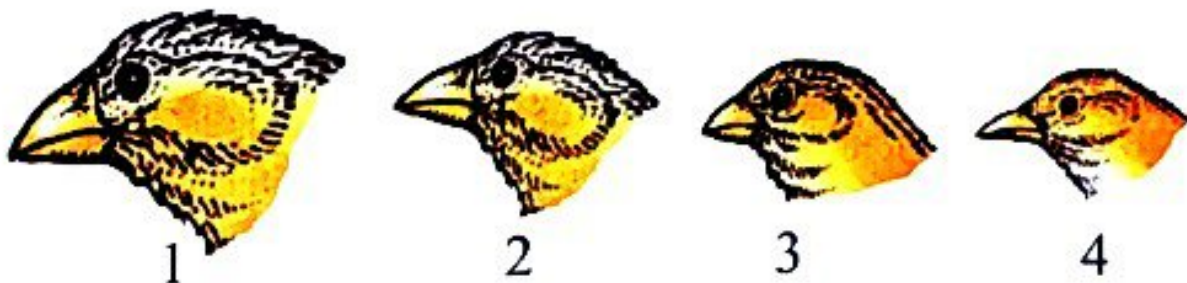
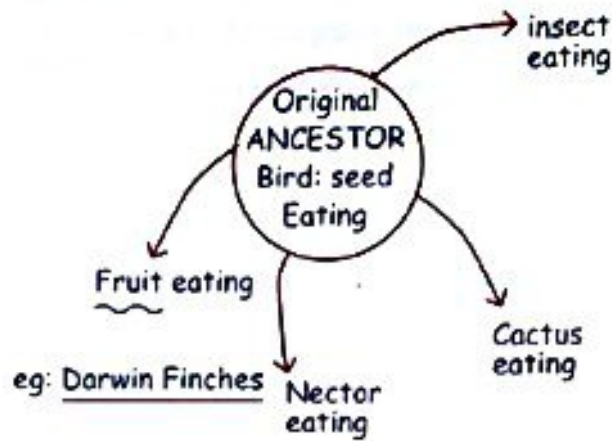


Fig: Variety of beaks of finches that Darwin found in Galapagos Island

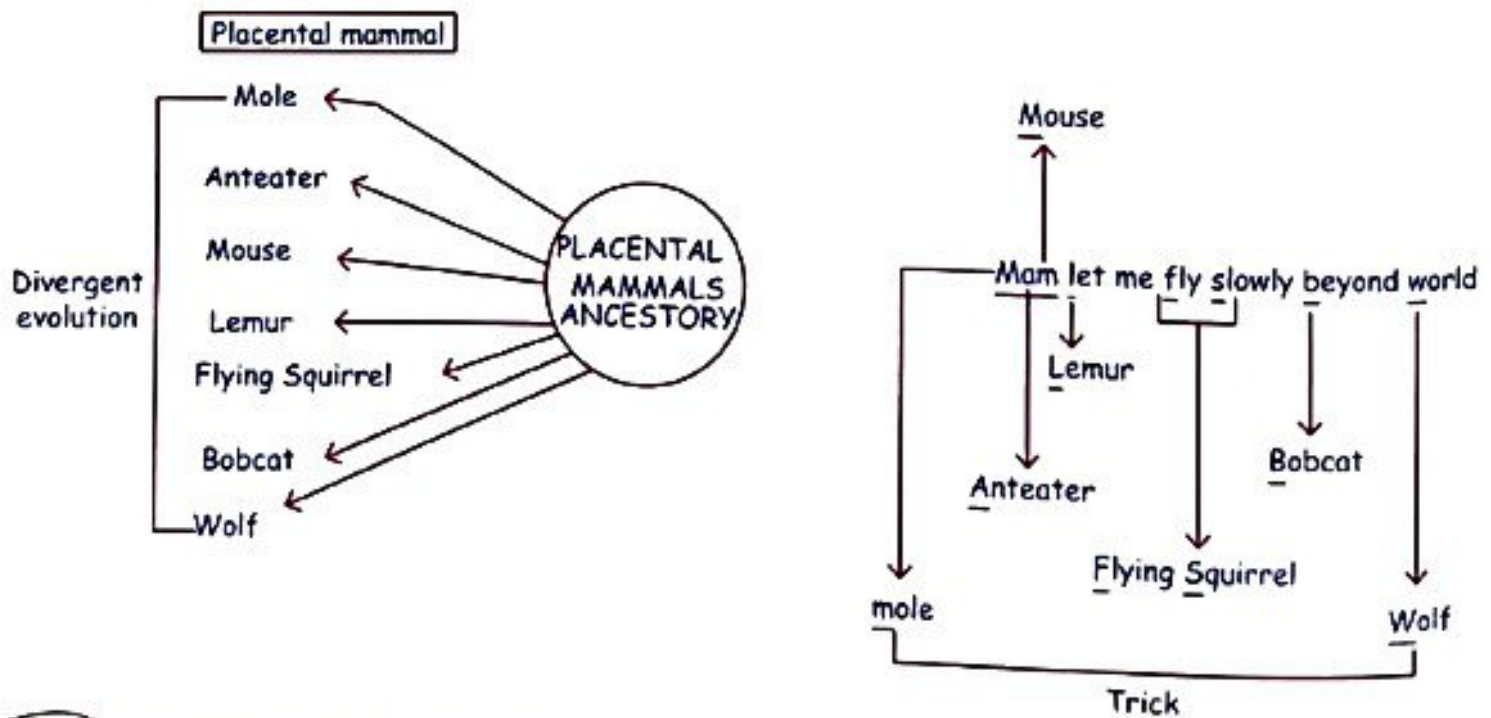


○ Beak modified as per feeding habit, type of food they fed upon.

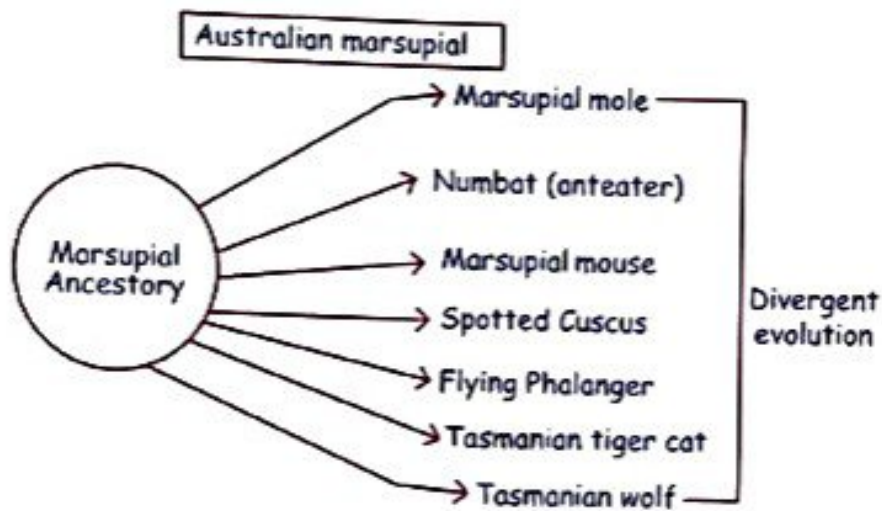
(ii) Australian Marsupial



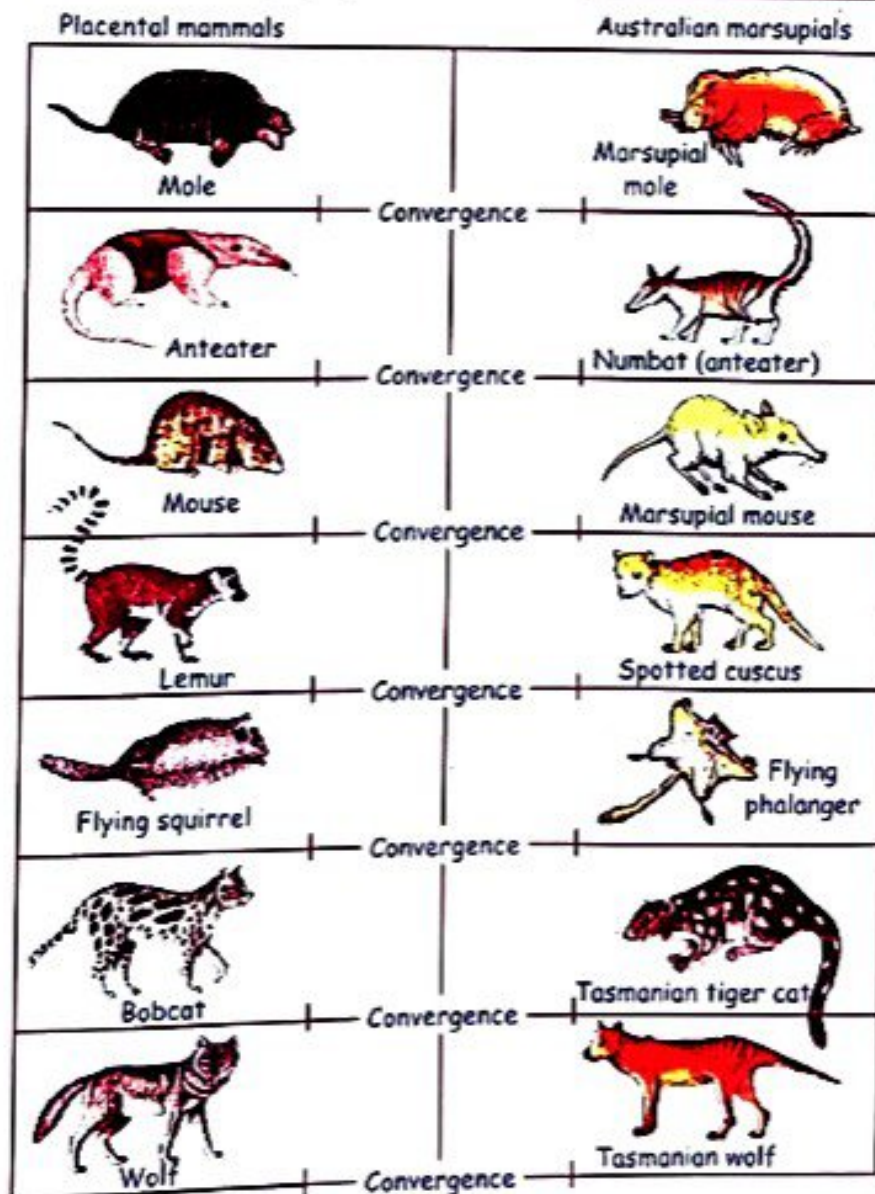
(iii) Placental mammals in Australia



NOTE Just remember these placental mammals, rest all e.g. are Australian marsupial

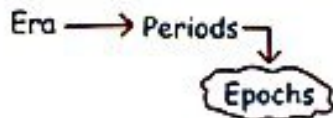


NOTE *Adaptive radiation of a given species indicates DIVERGENT EVOLUTION but if more than one adaptive radiation occurs in an isolated geographical area: It is CONVERGENT EVOLUTION.

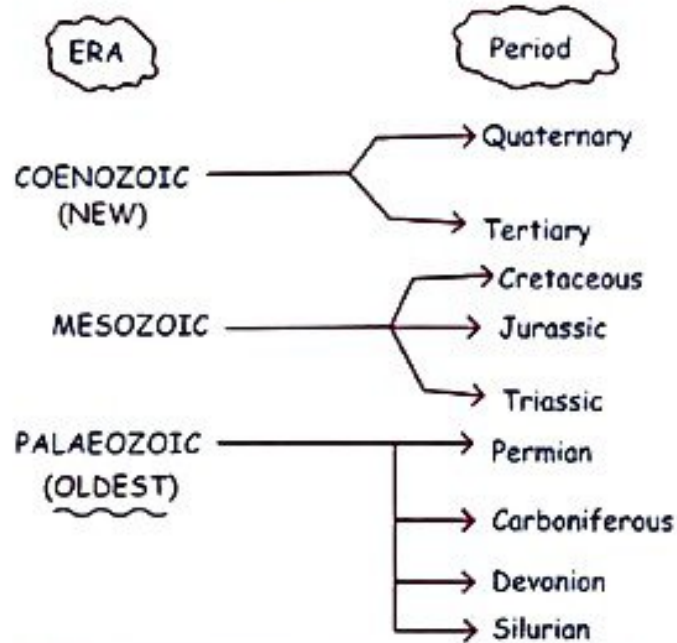


NOTE Members of placental mammals show convergence with Australian marsupials

GEOLOGICAL TIME SCALE

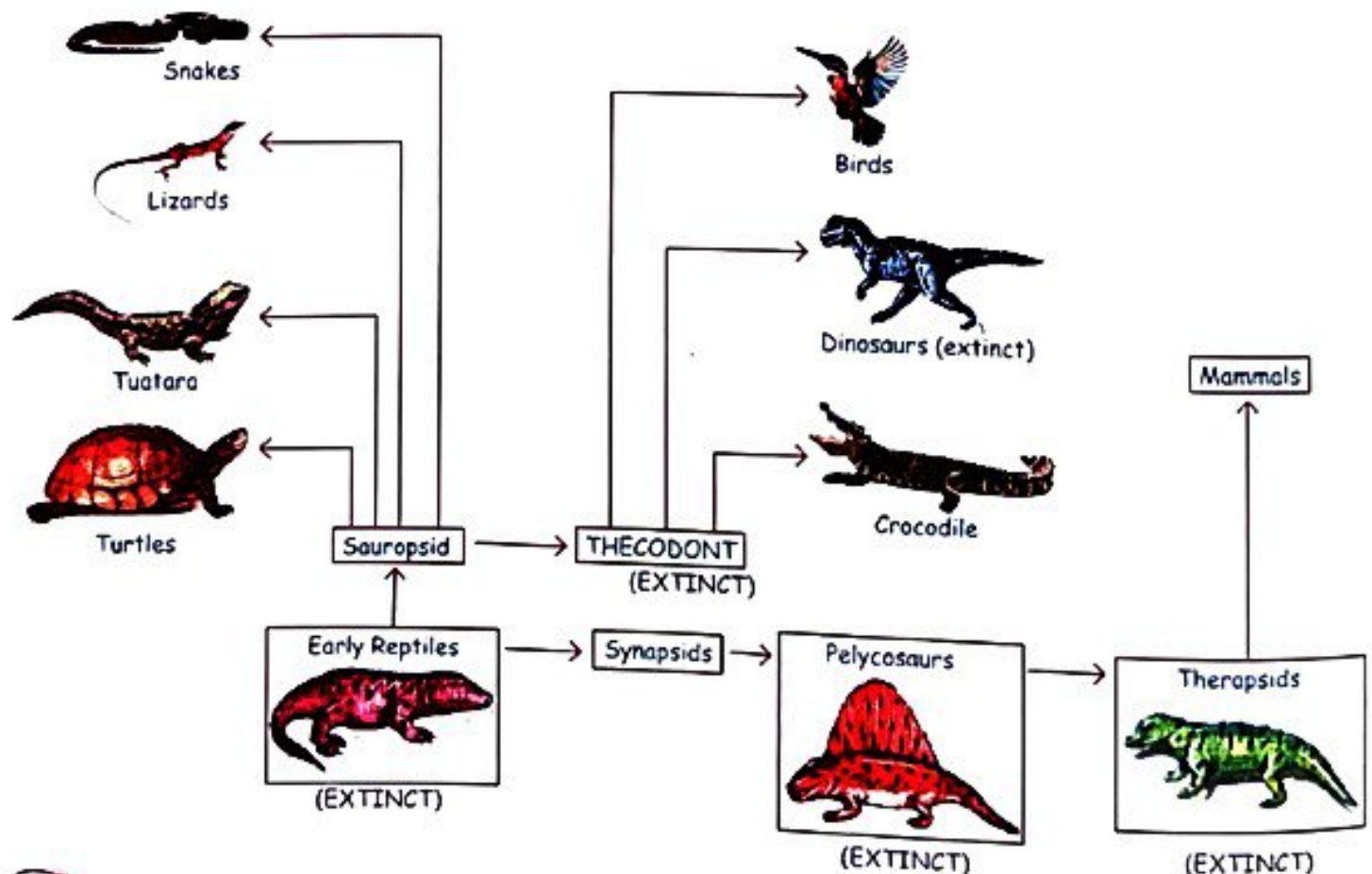


Era > Periods > Epochs

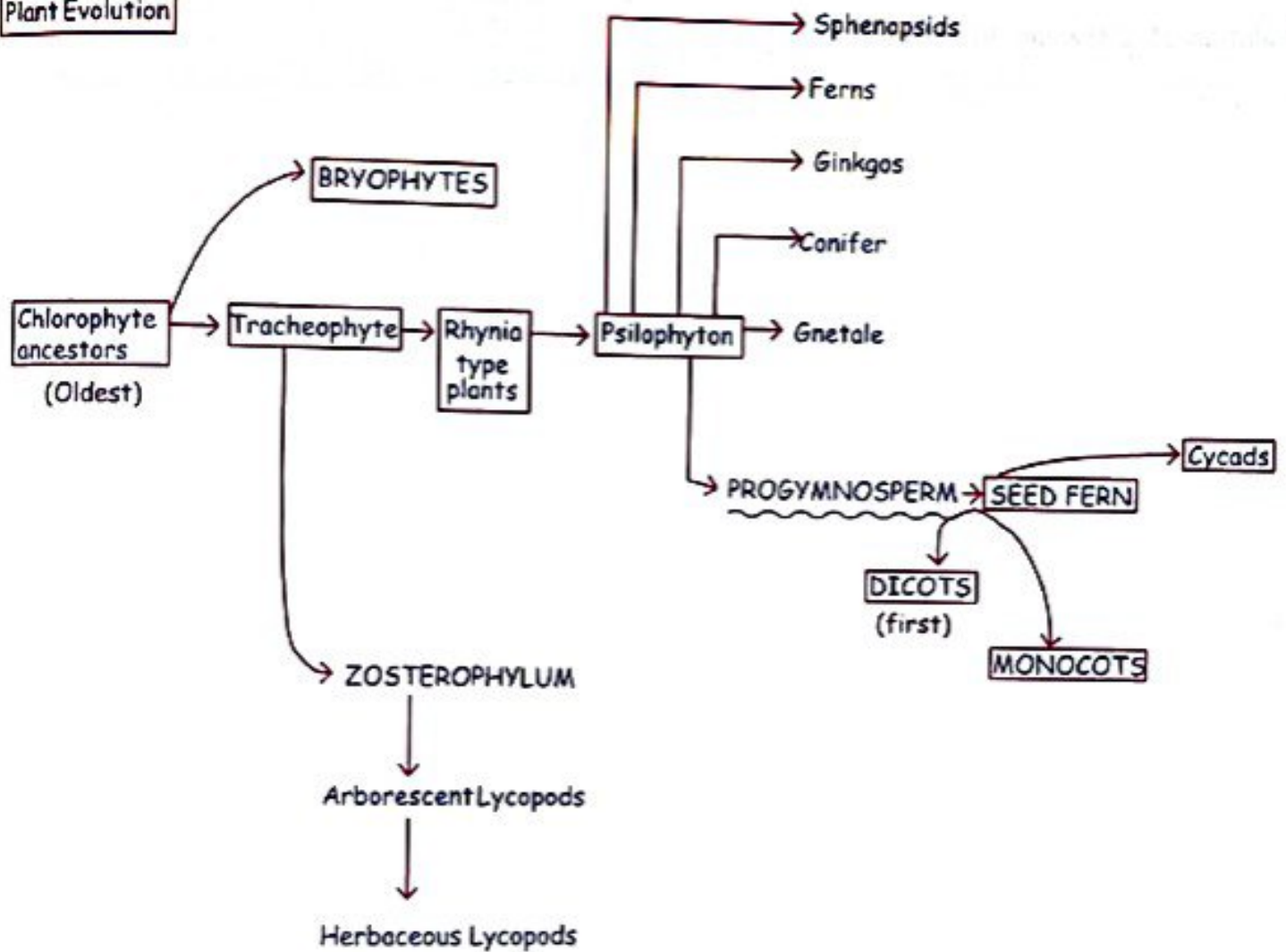


(Only as Per NCERT)

Evolutionary history of different vertebrates



Plant Evolution



- NOTE**
- ① Land Reptiles: Dinosaurs
 - ↓
 - Tyrannosaurus rex
 - ↓
 - Biggest Dinosaur of the that time (20 feet the height)
 - ↓
 - Huge fearsome, Dagger like Teeth
 - ② Initially there was a common land (PANGEA) but due to continental drift, South America joined North America. Initially South America had mammals resembling horse, hippopotamus, bear, rabbit etc but now these animal were overridden by North America fauna.
 - ③ Due to continental drift: pouched mammal/Australia marsupial survived in Australia due to lack of competition.

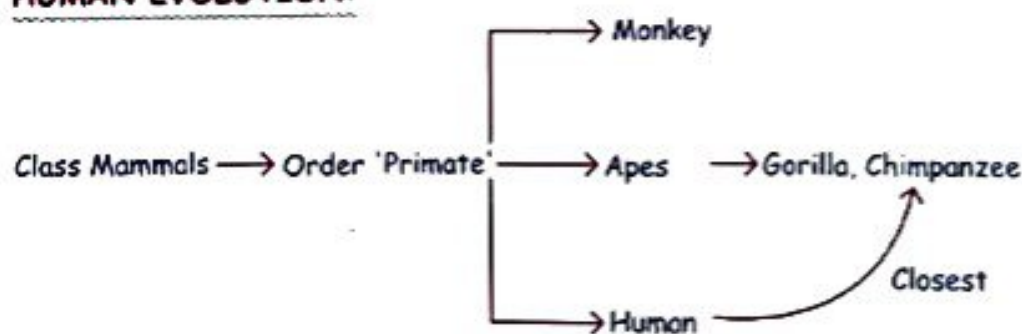
Evolution of different life forms:

- Origin of universe → 20 billion years ago
- Origin of Earth → 4.5 billion years ago
- Life appeared → 4 billion years ago (500 million years after formation of the Earth)
- Non cellular form of life → 3 billion years ago
- Cellular form of life → 2 billion years ago
- Invertebrates → 500 million years ago
- Jawless Fish → 350 million years ago
- Fish with stout and Strong fin could move to land & back to water might have → 350 million years ago evolved into amphibians.

(In 1938, an organism coelocanth/Lobe Fin fish was caught in South africa which was though to be extinct, is a connecting link between Fish & AMPHIBIAN)

- Plant & sea weeds → 320 million years ago
 - ↓ next 200 million years ago Reptiles dominated the earth
- Some land reptiles, moved to water & evolved into fish like reptiles (Ichthyosaurs) → 200 million years ago
- Dinosaurs disappeared (True Reason unknown) → 65 million years ago
- When reptiles came down mammals took over → First mammals were SHREW LIKE
 - Small fossils

HUMAN EVOLUTION:



Doctor	<u>Dryopithecus</u>
Ram	<u>Ramapithecus</u>
Australia Se	<u>Australopithecus</u>
H	<u>Homo habilis</u>
E	<u>Homo erectus</u>
N	<u>Neanderthal man</u>
लारे	<u>Cro-magnon (Not in NCERT)</u>
	<u>Modern man/Homo sapien sapien</u>

Doctor Ram Australia से H E N लारे
(TRICK)

Dryopithecus

- 15 million years ago
- Walked on 4 feet, hairy
- More ape like

Ramapithecus

(fossil obtained in SHIVALIK HILL)

India

- 15 million years ago
- Walked on 4 feet, hairy
- More Man like

Australopithecus

Lived on East african grassland, 2 million years ago

first time 'Bipedal movement'

(Cranial Capacity): 500 cc (Cubic cm)

Made stone weapon: fruit eating

Homo habilis

- Hominid/Human Like
- Made perfect tool
- Cranial capacity: 650 - 800 cc
- They did not eat meat probably.

Homo erectus

Cranial capacity: 900 cc

They had erect posture

Lived ~ 1.5 mya

The use fire for the first time

They ate meat

Fossil (Java man) was obtained in 1891

Neanderthal

Lived in Central & East Asia $\approx$ 1 lakh- 40000 year ago

Cranial capacity 1400 cc

For the first time covered their body, buried Dead one, put flower

Cro-magnon

- Cranial capacity: 1650 cc
- Made cave paintings (A cave painting was obtained, from Bhimbetka, Raisen District, Madhya pradesh, 18000 years ago)

Homo sapien sapien

Evolved during ICE AGE (75000 - 10000 yr ago)

Cranial capacity 1350 cc

10000 years ago Agriculture, human settlement began.

NOTE

- A few fossils were obtained from Tanzania & Ethopia of East Africa revealing human like features indicating hominid like creatures probably lived 3-4 mya.
- They were probably were not taller than 4 feet.

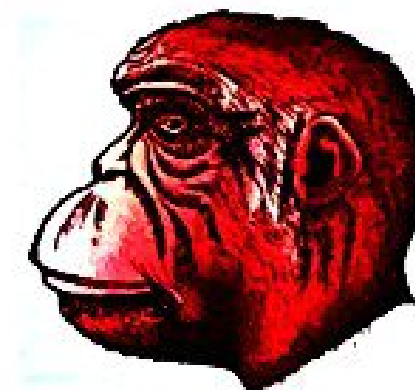
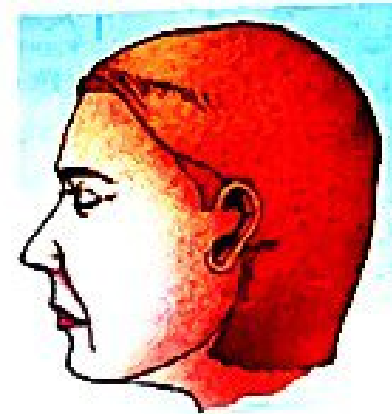


Fig. A comparison of the skulls of adult modern human being, baby chimpanzee and adult chimpanzee. The skull of baby chimpanzee is more like adult human skull than adult chimpanzee skull (Reveals about common ancestry)